ICT & DIGITAL INITIATIVES TRAINING PROGRAM

Connected Technologies:

Introduction to IoT and Smart Systems

Teacher Training Workshop | ~1 Hour Session

Session Objectives

1

Understand Internet of Things (IoT)

Learn what IoT means and why it matters in today's world

2

Learn how devices connect and communicate

Discover how smart devices talk to each other via internet

3

Explore smart systems and real-life applications

See IoT in action: homes, cities, healthcare & more

4

Discuss IoT in Education

How smart technology is transforming classrooms

What is IoT?

Internet of Things (IoT)

A network of physical objects — devices, vehicles, appliances — embedded with sensors, software, and connectivity that enables them to collect and exchange data over the internet.

Simple Explanation

Think of it like this:

Everyday objects (fridge, thermostat, streetlights) become 'smart' when connected to the internet — they can sense, think, and act on their own.

Everyday Example

Smart Home Thermostat

- Senses room temperature
- Learns your schedule
- Adjusts heating automatically
- You control it from your phone!

How IoT Works

A simple 5-step flow from device to user

1. Device

A physical object with sensors

2. Sensors

Detect temperature, light, motion, etc.

3. Internet

Data travels via WiFi / Bluetooth

4. Cloud

Data is stored & processed

5. You!

View & control via phone or computer

The whole process happens in seconds — automatically, 24/7!

Key Components of IoT

Sensors

Detect changes in the physical world (temperature, motion, light, pressure)

Microcontrollers

Tiny computers that process sensor data and make decisions

Connectivity

WiFi, Bluetooth, 4G/5G, Zigbee — how devices send data

Cloud Platforms

Remote servers that store, analyse and manage data from devices

Applications

Apps & dashboards where users view data and control devices

IoT Architecture

Four layers work together to make IoT systems function

Layer 1 — Device Layer

Physical 'things': sensors, actuators, cameras, thermostats

⇕

Layer 2 — Network Layer

How data moves: WiFi, Bluetooth, cellular networks

⇕

Layer 3 — Cloud / Data Layer

Servers process and store massive amounts of device data

⇕

Layer 4 — Application Layer

End user dashboards, apps, and automation rules

IoT in Daily Life

Smart Homes

Control lights, locks, AC, appliances from anywhere via phone

Smart Cities

Intelligent traffic lights, waste management, pollution monitoring

Healthcare

Remote patient monitoring, smart insulin pumps, hospital tracking

Smart Watches

Track your heartbeat, steps, sleep, and receive notifications

Smart Agriculture

Soil sensors, automated irrigation, drone monitoring of crops

IoT in Education

Technology is transforming how we teach and learn

Smart Classrooms

Interactive boards, automated projectors, smart lighting that adjusts for different lessons

Attendance Systems

RFID or face-recognition attendance — automatic, accurate, no paper registers

Environmental Monitoring

Sensors track temperature, air quality, and noise levels for healthy learning

Interactive Learning

Smart labs with sensor kits, robotics, and real-time data collection experiments

Smart Systems

Smart Systems = IoT + Artificial Intelligence (AI)

When devices not only collect data, but also learn from it and make decisions automatically

IoT Devices

Collect real-world data

+

AI / Machine Learning

Learns patterns & predicts

=

Smart System

Acts automatically, improves over time

Automation Examples:

- Smart AC that adjusts when it senses you're asleep
- Traffic system that changes signals based on congestion
- Factory robot that detects defects and sorts products

Benefits of IoT

Efficiency

Devices perform tasks faster, with less human effort and fewer errors

Automation

Systems work 24/7 without manual intervention — lights turn off when nobody's home

Data-Driven Decisions

Real-time data helps managers, doctors, and teachers make smarter choices

Convenience

Control everything from one app — home security, temperature, appliances

Challenges of IoT

Being aware of limitations helps us use IoT responsibly

Security

Connected devices can be hacked if not properly secured. Weak passwords and outdated software are common risks.

Privacy

Devices constantly collect personal data — usage patterns, location, health — raising privacy concerns.

Connectivity

IoT devices need reliable internet. Poor networks in rural areas limit adoption and functionality.

Cost

Installing smart systems can be expensive. Maintenance and upgrades add to long-term costs.

IoT Security Basics

⚠ **Security is everyone's responsibility — teachers included!**

01

Use Strong Passwords

Change default device passwords immediately. Use long, unique passwords for every device and account.

✗ *admin123* → ✓ *Tr@!nSmart#2024*

02

Secure Your Network

Use WPA3 encryption on your WiFi. Create a separate guest network for IoT devices.

Keep IoT devices on a separate WiFi network

03

Apply Regular Updates

Update firmware and software when prompted. Updates fix security vulnerabilities.

Enable automatic updates where possible

Hands-On Activity

Try It Yourself! — Smart Temperature & Light Demo

What You Need

- Arduino/ ESP 8266 / Raspberry Pi kit (or simulation)
- Temperature sensor (DHT11)
- LED light module
- Laptop with USB connection
- Free Tinkercad or Wokwi simulator

Activity Steps

1. Connect temperature & humidity sensor (DHT-11/22) to board
2. Upload simple code to read temperature/humidity
3. Program LED to turn ON if temp > 30°C
4. Monitor readings on screen in real-time
5. Discuss: How would this work in a classroom?

The Future of IoT

By 2030, over 30 billion IoT devices are expected to be connected worldwide

Smart Cities

Entire cities managed by IoT: smart traffic, energy grids, emergency response, and waste systems

Industry 4.0

Factories run by intelligent machines — predictive maintenance, automated logistics, zero downtime

AI + IoT

AI analyses IoT data instantly. Self-driving cars, predictive healthcare, personalised learning

Future Education

Smart campuses, AI tutors, IoT labs in every school — preparing students for a connected world

Conclusion & Discussion

Key Takeaways

- IoT connects everyday devices to the internet
- Smart systems combine IoT + AI for automation
- IoT is already in our homes, cities, and schools
- Security & privacy must be taken seriously
- The future is connected — and it starts in our classrooms!

Discussion Questions

Q1 *Where do you currently see IoT in your daily life?*

Q2 *How could smart technology improve your classroom?*

Q3 *What concerns do you have about IoT security?*

Q4 *What would an ideal IoT-enabled school look like?*